

Line Measurement

Baker College Engineering

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College of Information Technology and Engineering

Agenda

Instrument Selection

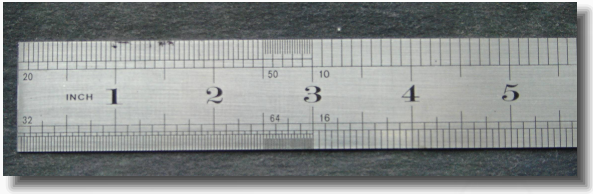
Line Measurement

Instrument Selection



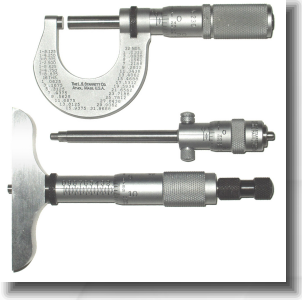
General Groups

Non-precision line-graduated measuring instruments



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Precision Measuring Instruments



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Non-Precision vs Precision

Non-precision line-graduated

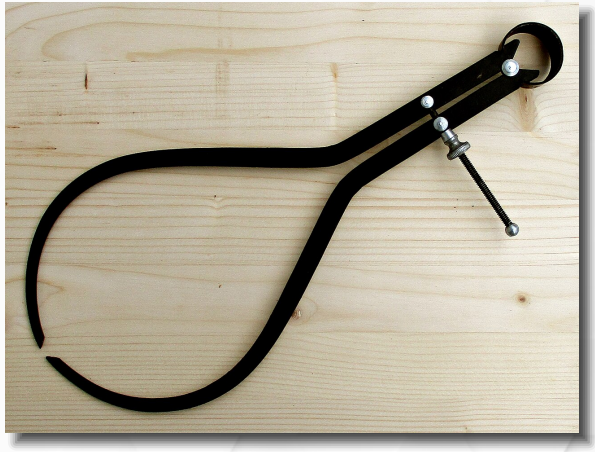
- Accuracy is highly dependent on the user
- Capable in applications not requiring precision in thousandths

Precision

- Accuracy is dependent on the sensitivity/accuracy of the instrument
- Capable down to tenths of thousandths (or smaller)

Types of Rules

1. Rules
2. Calipers (→)
3. Surface gages
4. Depth gages



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These are *calipers*–they are set to the size of the object being measured, locked in place with the screw, then measured with a *rule*

Benefits of Non-precision

- Temperature doesn't matter as much–graduations are usually bigger than any growth/shrinkage
- Measurment pressure doesn't matter–did you screw down the micrometer too hard?

Selecting the Proper Instrument

1. Physical characteristics of the part
2. Dimensions to be measured
3. Accuracy required
4. Convenience
5. Adaptability
6. Speed
7. Reliability

Accuracy of Measurement (1 of 2)

What is the smallest unit you can measure?

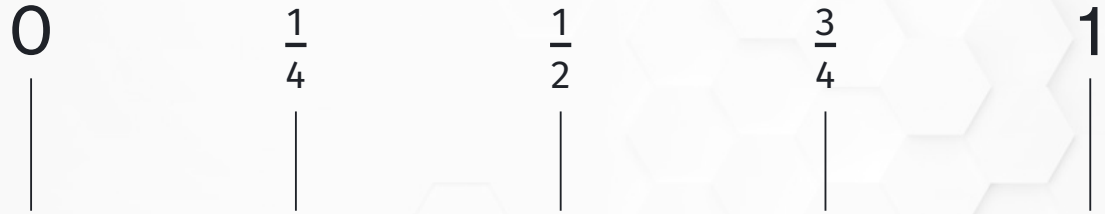
Accuracy of Measurement (2 of 2)

No instrument is capable of controlling the accuracy of measurement finer than the finest graduation on its scale

Control vs Estimation

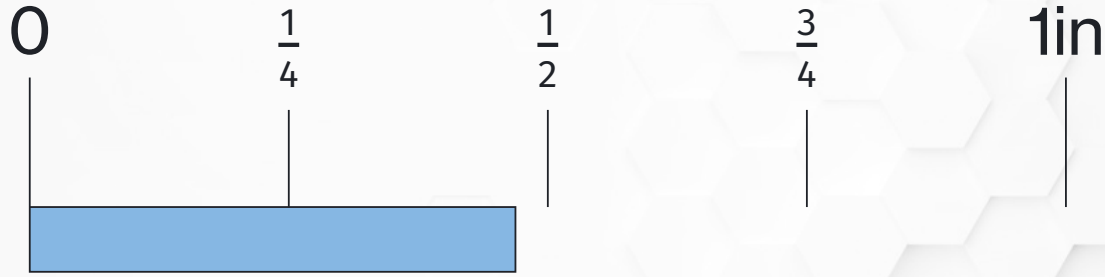
- You can only *control* a measurement as small as the finest graduation
- You can *estimate* a measurement half the size of the finest graduation

Accuracy Example (1 of 4)



Let's use this example for our measurement–a scale that reads from 0 to 1 with divisions at each quarter.

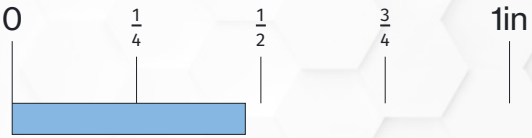
Accuracy Example (2 of 4)



What is the measurement of the blue bar?

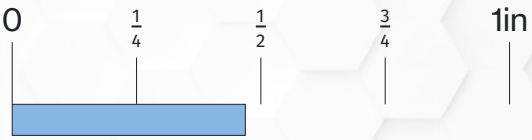
Accuracy Example (3 of 4)

- We can only *control* the graduations
- It is either 1/4 or 1/2

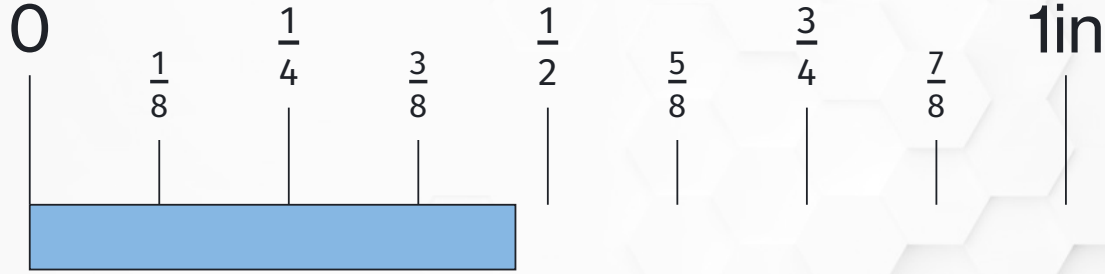


Accuracy Example (4 of 4)

- We can *estimate* 1/2 the graduations
- We can estimate 3/8 or 1/2



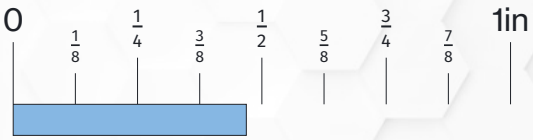
Increasing Accuracy (1 of 3)



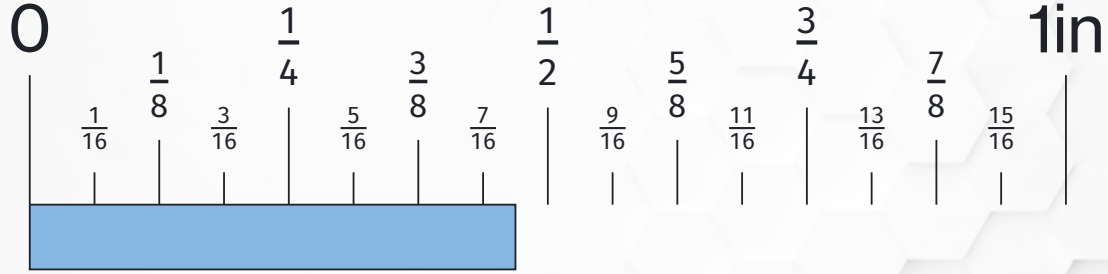
Adding divisions allows us to *control* smaller dimensions, increasing the measurement accuracy

Increasing Accuracy (2 of 3)

- Now we can *control* 3/8 or 1/2
- Now we can *estimate* 7/16 or 1/2

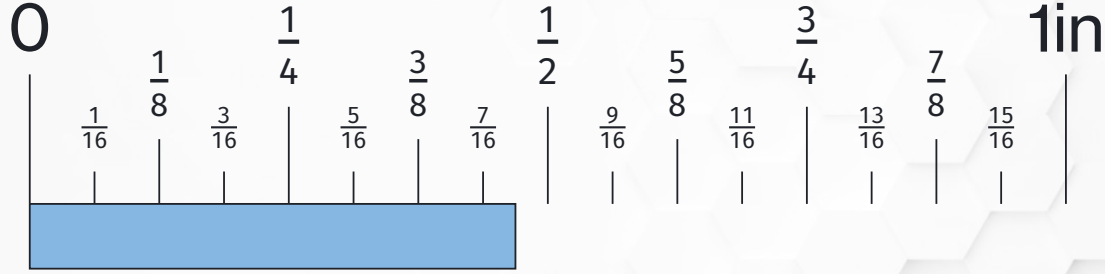


Increasing Accuracy (3 of 3)



We can continue adding graduations–with 1/16 graduations we can *control* 7/16 or 1/2 and we can *estimate* 15/32

Rules–Accuracy Limit



The drawback is that eventually we'll end up with graduations too small to see–that's where we switch to precision instruments

Line Measurement



Line Measurement

Rules

Rules



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Traditional Rules

- Traditional rulers are made of wood
- Prone to damage
- Damaged edge = decreased accuracy



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Line Measurement

Steel Rules

Steel Rules (1 of 2)



Steel Rules (2 of 2)

- Markings etched into steel – more accurate
- Steel is less likely to become damaged



Steel Rule Variety

- Available in many different sizes and graduations
- Always pick the proper graduation for the job



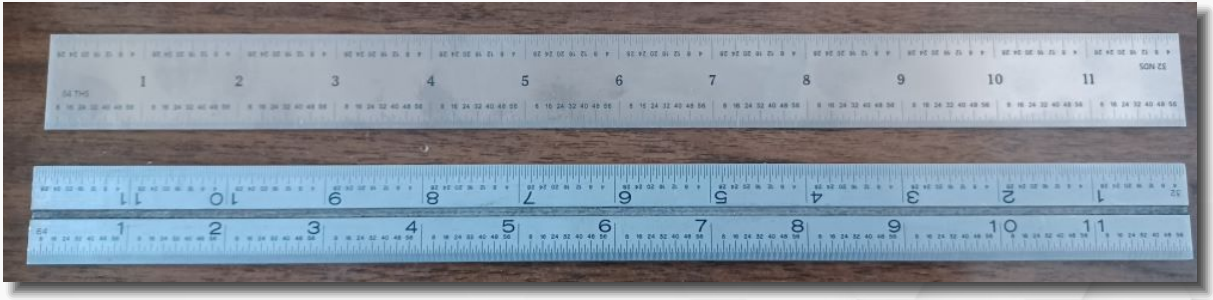
Line Measurement

Tempered Steel Rules



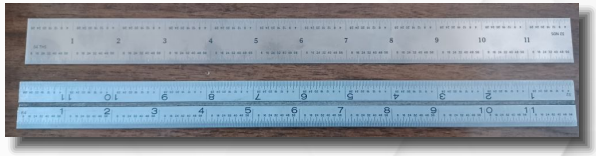
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Tempered Steel Rules (1 of 2)



Tempered Steel Rules (2 of 2)

- Available in many different sizes/graduations
- Typically VERY stiff – not meant to be bent
- Some include a center channel for accessory clamps

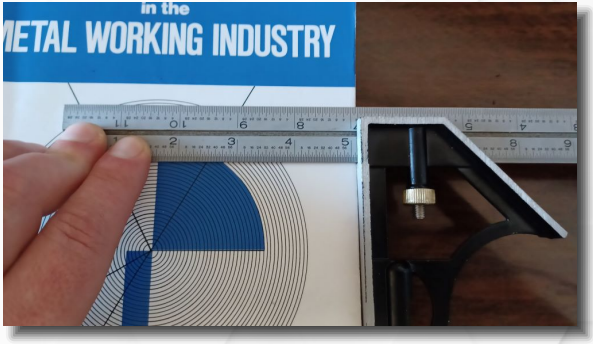


Tempered Steel Rule Accessories



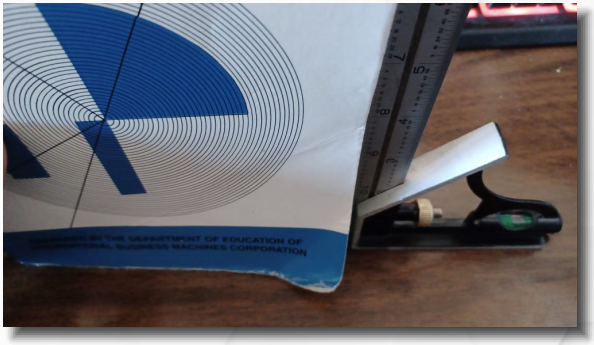
Square Clamp

- Allows for precise measurement at a right angle
- Usually has a fixed 45 degree angle
- Usually has a spirit level



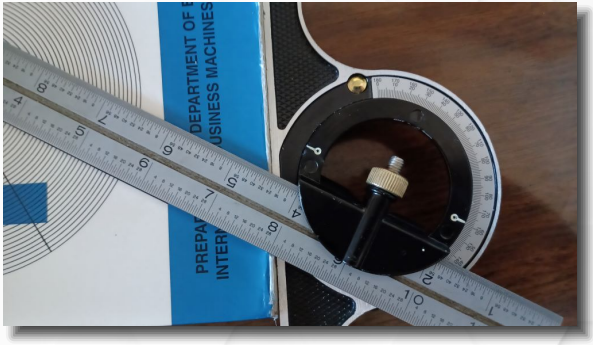
Height Gauge

- Moved to the end of the rule it can also be used as a height gauge



Protractor Clamp

- Used to measure angles
- Usually includes a spirit level in line with the *rule*, not the protractor flat



Center Clamp

- Used to locate the center of a circular part



Line Measurement

Machinist Scales

Machinist Scales (1 of 2)

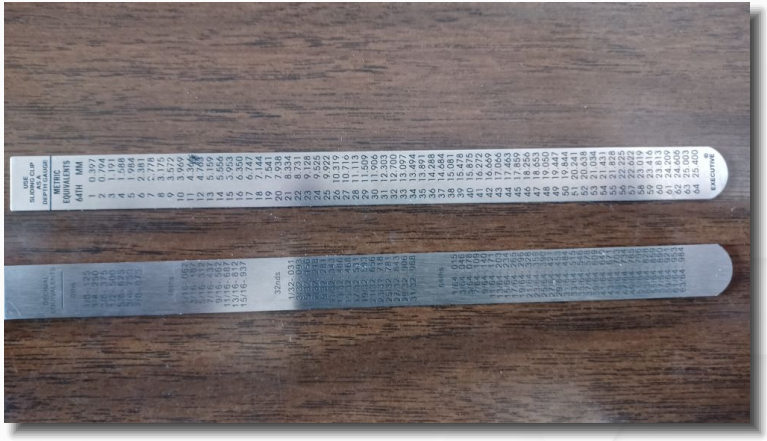


Machinist Scales (2 of 2)

- Typically 6” length
- One end rounded
- Typically more than one graduation scale
  - 32nds are most common
  - Many have 64ths and/or metric



Reference Material



The scale is usually only on one side of the machinist scale. The other side typically has some sort of reference material–fraction to decimal is common, but it can also have unit conversion, etc.

Height/Depth Gauge

- Includes pocket clip
- One end is rounded – keeps corners from stabbing you
- BUT – the clip also makes a quick height/depth gauge



Steel Rule Video

The Steel Rule

Questions?

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